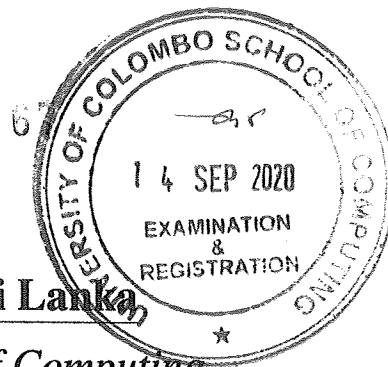




University of Colombo, Sri Lanka

University of Colombo School of Computing

Bachelor of Science in Computer Science

Bachelor of Science Honours in Computer Science

Bachelor of Science Honours in Software Engineering

First Year Examination — Semester I - 2020

SCS1206 — Laboratory I (Part A)

Two (2) Hours (for both Parts A & B)

Answer All Questions

Number of Pages = 5

Number of Questions = 2

To be completed by the candidate

Index Number

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Important Instructions to candidates

- The students are required to answer both **Part A** and **Part B** in two hours.
- The medium of instructions and questions is English.
- You shall write your answers in **English**.
- This paper has **2** questions on **5** pages.
- Answer **all 2** questions.
- **Write your answers on and only on the space provided** on this question paper.
- Write your index number on each and every page of the question paper.
- Do not tear off any part of this answer book. Under no circumstances may this book (or any part of this book), used or unused, be removed from the Examination Hall by a candidate.
- Questions appear on both sides of the paper. If a page is not printed, please inform the supervisor immediately.
- Any electronic device capable of storing and retrieving text, including electronic dictionaries and mobile phones, are **not allowed**.
- Non-programmable Calculators may be used.

To be completed by the examiners

1	
2	
Total	

--	--	--	--	--	--	--	--

- [4 marks]**



- [1 marks]**

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- [1 marks]**

- [1 marks]

[illegible]

- [2 marks]

- [2 marks]

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- [2 marks]**

The device file `sda` denotes a _____ whereas `sda1` denotes a _____.

Index Number

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(d). Everything in GNU/Linux is a file.

- i. Assume you log into a Ubuntu GNU/Linux operating system remotely through a shell. Assume, you are at `/home/kenneth/documents/backup/`. What is the **relative path** from your current location to the file

`/home/kenneth/downloads/tute/intro.pdf`

[2 marks]

- ii. If you are in the directory same as the file `intro.pdf`, what is the shell command that can be used to find the size of `intro.pdf` file?

[2 marks]

- iii. The result of `ls -l intro.pdf` is given below.

```
-rwxr-x--- 1 kenneth kenneth 67235 May 31 2019 intro.pdf
```

Explain the permissions of the file `intro.pdf` using given result.

[8 marks]

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--	--	--	--	--	--	--	--

- i. Complete the given $\text{\LaTeX}$ script correctly to generate a valid output.

... { document }

- [3 marks]

- i. List two (2) advantages of source code version management.

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Index Number

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- ii. What is the git command used to create a local repository in current directory?

[2 marks]

- iii. What is the git command used to replace a file in local repository with a file from staging area?

[2 marks]

- (c). Text based editors such as Vim are very useful when using command line interface.

- i. List two (2) alternative text based editors to Vim that can be used in GNU/Linux.

[2 marks]

--

- ii. What is the Vim command syntax used to search and replace content without prompting the user?

[3 marks]

- iii. What is the Vim command syntax used to map a shell command to a keyboard key combination?

[3 marks]



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Engineering**

First Year Examination — Semester I - 2020

SCS1206 — Laboratory I (Part B)

Two (2) Hours (For both part A and B)

Answer All Questions

Number of Pages = 8

Number of Questions = 2

To be completed by the candidate

Index Number

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Important Instructions to Candidates

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- Write your index number on each and every page of the question paper

To be completed
by the examiners

3	
4	
Total	

3. Consider the following writing compiled using libreoffice Writer.

[25 Marks]

Introduction to Computers

A computer is a man-made, programmable electronic machine that can store, retrieve, and process data according to a given set of instructions. Whatever information given to a computer is termed **input**. The computer input typically comprises of two things: **Data** and **Instructions**. The data are the facts, which are to be processed, and the instructions describe how the computer should process the data. The data which has been converted or organized into a more useful or intelligible form for direct utilisation or helps human beings in their decision making process is typically termed **information**.

Characteristics of Computers

The main characteristics of computers are:

- ❖ Speed
- ❖ Accuracy
- ❖ Diligence
- ❖ Versatility
- ❖ Power of Remembering
- ❖ No I. Q.
- ❖ No Feelings

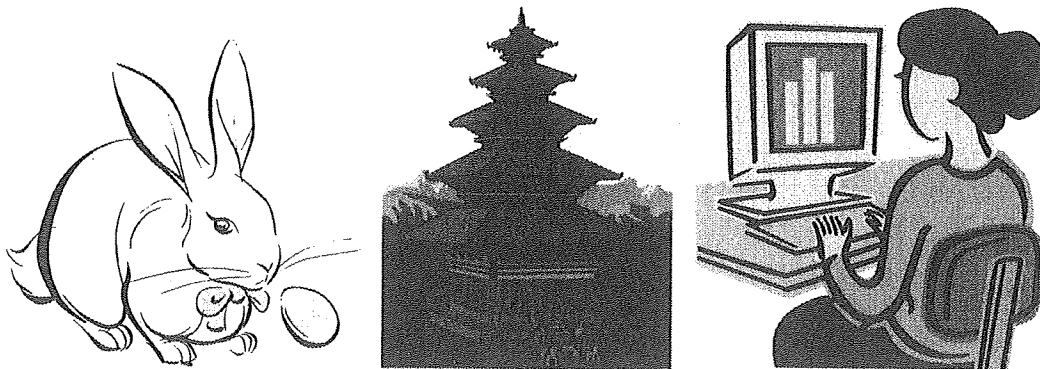
GENERATIONS OF COMPUTERS				
	FIRST (early 50's)	SECOND (late 50's)	THIRD (60's)	FOURTH (70's +)
Technology	vacuum tubes	transistors	Integrated circuits	microchips (LSI, VLSI)
Relative speed	1	100	100	1,000,000 +++
Relative cost	1	0.1	0.01	0.001 ---
RAM capacity (bytes)	1,000-8,000	4,000-64,000	32,000-4,000,000	32,000,000+++
External storage	cards	tape	magnetic disks	mass storage
Operating System	Single user; jobs scheduled manually	Single user; jobs scheduled automatically	Multiple users; time- sharing	Multiple users; distributed systems

Storage

Smallest Unit	- 1 bit
8 bits	- 1 byte
1024 bytes	- 1 kilobyte (KB)
1024 kilobytes	- 1 megabyte (MB)
1024 megabytes	- 1 gigabyte (GB)
1024 gigabytes	- 1 terabyte

(In computer jargon, One Kilo = 2^{10} = 1024)

Inserting a picture:



Explain cut, copy, and paste.

The **Cut**, **Copy**, and **Paste** commands are nearly universal. These three functions are used by almost every Windows program and perform more or less the same function in each of them.

Cut: When you cut something, it is deleted from its current location and saved to the Clipboard.

Copy: Copy is similar to Cut except the original item is not deleted.

Paste: You can place information on the Clipboard wherever you like. Execute the Paste command and information you have cut or copied is placed wherever your cursor is located.

Clipboard: The Clipboard is the storage area for items that have been cut or copied.

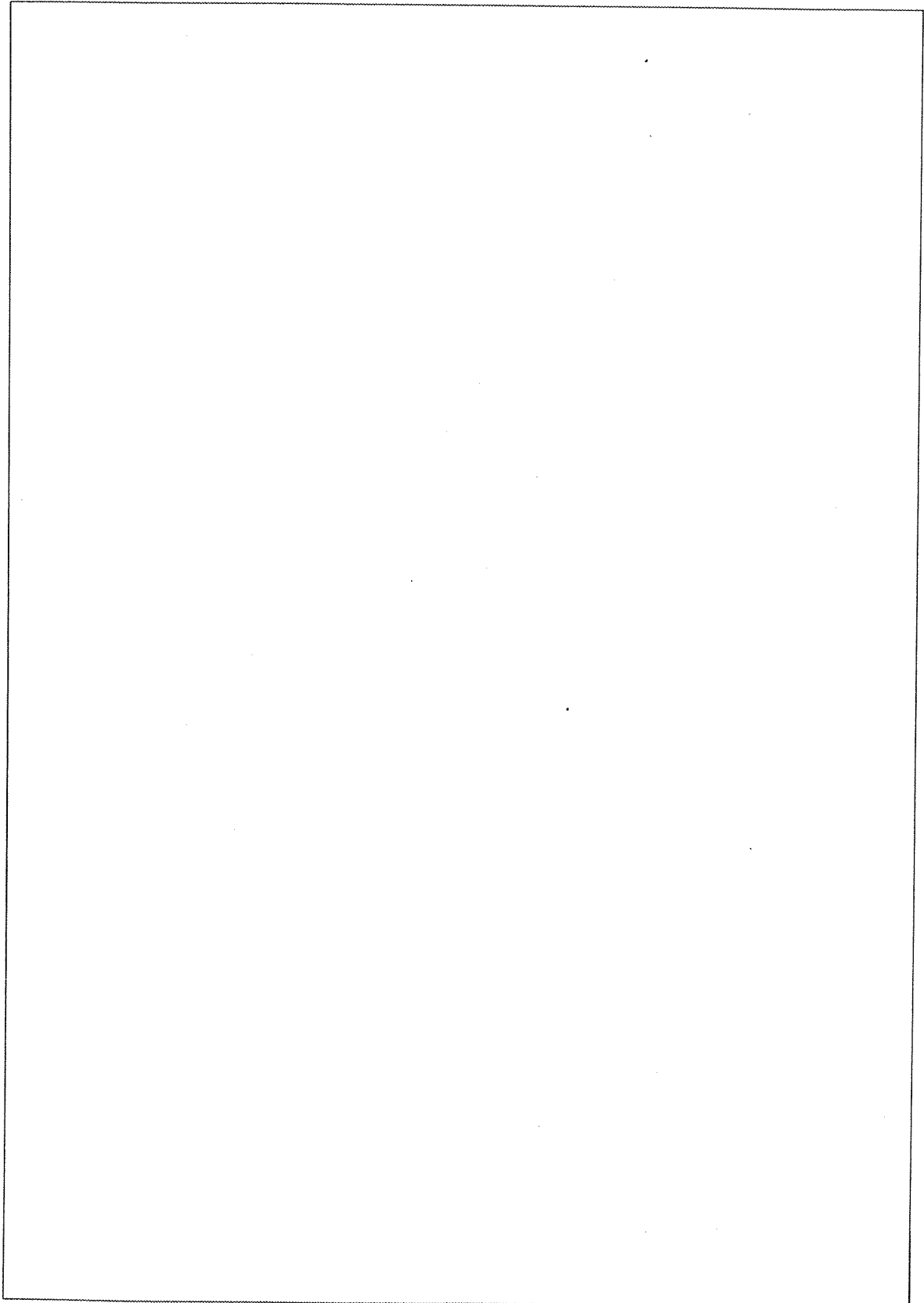
Instructions to follow up :

All the text must be typed as it is on this document.

Page Setup :

Paper Size : A4 , Portrait
Margins Top : 2 cm Bottom : 2.2 cm Left : 2.5 cm Right : 2 cm

3.1. Write in point form all the facilities used to format the document that are available in the libreoffice.



3.2. Describe in nutshell the usage of libreoffice writer for a 1. Computer science Student 2. Project Manager 3. Quality assurance officer 4. Team lead who are working at a small software engineering company

1.

2.

3.

4.

4. Consider the following Libreoffice calc work sheet.

[25 Marks]

	A	B	C	D	E	F	G	H	I	J	K	L
1												
2												
3												
4	Index No	First Name	Last Name	Subject 1	Subject 2	Subject 3	Subject 4	Total	Average	Result	Rank	
5	1001	Sunil	Liyanage	65	78	56	45	?	?	?	?	
6	1002	Nimal	Ratnayake	53	85	88	78	?	?	?	?	
7	1003	Sarath	Godage	45	96	ab	98	?	?	?	?	
8	1004	Kamal	Silva	78	76	76	65	?	?	?	?	
9	1005	Dhanuka	Withanage	23	ab	74	36	?	?	?	?	
10	1006	Lalith	Mendis	78	32	64	52	?	?	?	?	
11	1007	Kamala	Peiris	89	65	35	41	?	?	?	?	
12	1008	Tharumini	Withanage	65	94	62	74	?	?	?	?	
13	1009	Seetha	Watareka	88	78	61	ab	?	?	?	?	
14	1010	Neela	Liyanage	32	55	81	95	?	?	?	?	
15												
16	Maximum			?	?	?	?	?	?			
17	Minimum			?	?	?	?	?	?			
18												
19	No. of Students		??									
20	No. of Attendance			?	?	?	?					
21	No. of Absentee			?	?	?	?					
22												

4.1. Write down all the formulae required to calculate the following by explaining input required.

4.1.1.Total

4.1.2.Average

4.1.3.Result → Pass if average is greater than to 65 else fail

4.1.4.Rank

4.1.5.Maximum

4.1.6.Minimum

4.1.7.No of students

4.1.8.No of attendance

4.1.9.No of absence

4.1.1. Total

4.1.2. Average

4.1.3. Result → Pass if average is greater than to 65 else fail

4.1.4. Rank

4.1.5. Maximum

4.1.6. Minimum

4.1.7. No of students

4.1.8. No of attendance

4.1.9. No of absence